Deletion Scars: Inferring Deleted Artist Styles from Paired Diffusion Models

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Abstract. Concept erasure edits text-to-image diffusion models to suppress named visual concepts, including artist styles. Standard evaluations ask whether the target is suppressed and whether unrelated utility is preserved. We ask instead whether the edit reveals *which* concept was erased. Given query access to an original and an erased checkpoint and a closed candidate set, an auditor probes both models with matched prompts and seeds and measures displacement in DINOv2 space. On a frozen blinded holdout of ten untouched artist deletions (SD1.5, ESD-x), this paired displacement identifies the erased target far above chance: top-1 accuracy is 50% against a chance rate of 10%, and mean reciprocal rank is 0.576 against a chance value of 0.293, exhausting an exact 10! permutation test ($p = 1/10!$, the smallest attainable value). Raw displacement is nonetheless confounded: some candidates move under many unrelated erasures and become systematic false positives. We estimate this susceptibility from disjoint unrelated erasures and subtract it, leaving a residual excess we call a *deletion scar*. Calibration raises top-1 accuracy to 70% and MRR to 0.814 on the same holdout; the direction is consistent, but with ten targets the paired improvement over raw displacement is not established at conventional confidence (+0.238 MRR, bootstrap interval $[-0.088, +0.540]$). The underlying signal persists across five frozen prompt templates and reappears, at smaller scale, on SDXL and under Unified Concept Editing.

Keywords: Concept erasure · Diffusion models · Deletion inference · Model auditing

1 Introduction

Text-to-image diffusion models reproduce distinctive visual concepts from short natural-language prompts. This has motivated a body of work on post-hoc *concept erasure*: given a pretrained model and a target such as an artist style, the goal is to edit the model so the target is no longer produced on demand. Methods such as ESD [4] and Unified Concept Editing [5] alter parameters without retraining, and are evaluated by measuring target suppression, image quality, and preservation of unrelated concepts. Such evaluations begin from an implicit assumption: the identity of the erased concept is already known.

We study the complementary problem. An auditor receives two checkpoints, an original model M and an edited model M_t from which one unknown concept t has been erased, and knows that t belongs to a closed candidate set $\mathcal{C}$ but not

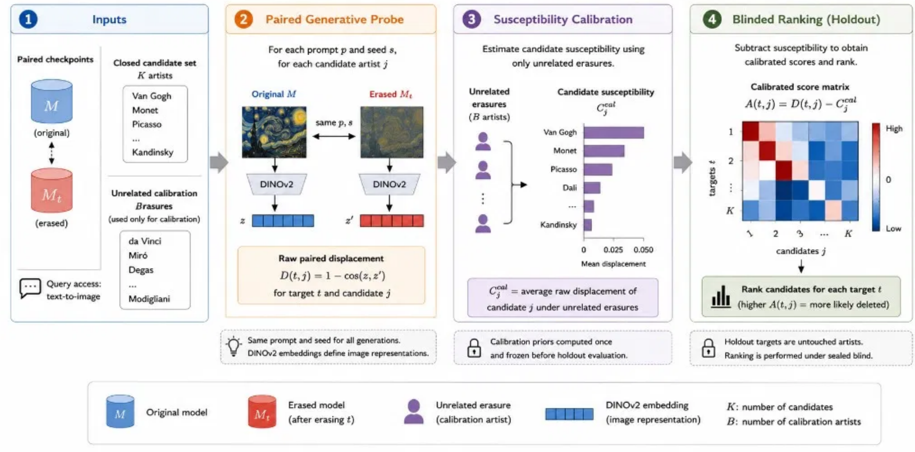

Fig. 1: The deletion-scar attack (Setting P). Given paired checkpoints, a closed candidate set, and B unrelated calibration erasures, matched prompts and seeds produce images from both models, embedded with DINOv2 to give the raw paired displacement $D(t, j)$. Calibration erasures estimate each candidate's background displacement $\widehat{C}_j^{\text{cal}}$, frozen before holdout evaluation; subtracting it gives $A(t, j) = D(t, j) - \widehat{C}_j^{\text{cal}}$, which ranks candidates for each sealed target.

which candidate was targeted. The auditor may sample from both checkpoints but does not inspect their parameters. Can the auditor determine what changed?

The task looks trivial at first: query every candidate on both checkpoints and pick the one whose outputs changed most. Our experiments show why that intuition is incomplete. Target-conditioned prompts do tend to change more than unrelated ones, but a confound sits on top of the signal. Some candidate prompts move substantially under *many* different erasures, whatever artist was actually targeted. In our discovery set Van Gogh behaves as a strong "scar attractor"; on the blinded holdout, Georges Seurat becomes a frequent false positive. The largest observed change therefore mixes a deletion-specific response with a candidate-specific tendency to move under editing. The useful question is not which candidate changed the most, but which candidate changed more than we would expect it to under an unrelated erasure.

We address the confound with *independent susceptibility calibration* (Fig. 1). Before evaluating the unknown deletion, we use a disjoint collection of unrelated artist-erasure checkpoints. For every candidate c_j we measure how much generations conditioned on c_j typically move when *some other* concept is erased, which gives a candidate-specific susceptibility prior. Subtracting that prior from the raw displacement leaves excess paired displacement: change that background susceptibility does not explain. We call this target-specific excess a *deletion scar*.

The attack, operating strength, encoder, candidate set, and calibration procedure were frozen before we evaluated ten previously untouched artist deletions, and predictions for all ten models were sealed before labels were revealed. Two claims follow, supported to different degrees. That erasure leaves recoverable target information in paired generative behavior is established: raw displacement alone ranks half the sealed targets first and exhausts an exact permutation test. That calibration improves the recovery is a description of this holdout rather than an established margin, since with ten targets the paired improvement carries a bootstrap interval containing zero (Sec. 5.3). We report both, and we report where calibration fails.

Prior work already establishes that model updates and unlearning can reveal what was removed [6, 7], so deletion-target inference is not a new problem. Our scope is narrower: how deletion-target information appears in the *paired generative behavior* of diffusion models, where the observable is a structured change in generated images and where candidate prompts differ sharply in their sensitivity to unrelated edits. Our contributions are:

1. **Generative deletion scars.** We formulate deletion-target inference for paired diffusion checkpoints under query-only access, and show that erasure produces target-specific excess displacement in image-embedding space.
2. **Independent susceptibility calibration.** We identify candidate susceptibility as the dominant confound in raw displacement, estimate it from unrelated erasures alone, and raise blinded holdout top-1 accuracy from 50% to 70%.
3. **Frozen blinded validation.** We evaluate ten untouched artist deletions under a protocol fixed before label reveal, and report where calibration fails and what the evidence does not establish.
4. **Replication and cost.** We recover the phenomenon on SDXL and under UCE at smaller scale, and characterize how much auxiliary access calibration needs.

The lesson is that the most informative change is not the largest change. To identify what was erased, one first has to know how unrelated erasures move the candidate space.

2 Related Work

Concept erasure in diffusion models. Post-hoc erasure suppresses selected concepts without full retraining. ESD modifies diffusion parameters with negative guidance [4], UCE performs closed-form cross-attention updates [5], and later work scales to mass removal [9]. These methods are evaluated by whether the target remains generatable and whether unrelated utility is preserved. A parallel auditing line shows that erasure is often incomplete: adversarial prompt search recovers nominally erased concepts from the edited model alone [3, 17, 18], textual inversion re-elicits them from the remaining parameters [11], latent traces persist [15], and editing one concept perturbs others [16].

Our question differs from recovery. We do not make the erased model reproduce the target; we ask whether the pattern of behavioral change between the two checkpoints identifies which candidate was targeted. The two properties come apart in both directions: a concept can be hard to generate and still leave no uniquely identifiable trace, while a target-specific paired trace can persist even when a suppression proxy barely moves, as our weak-CLIP cases show.

Deletion inference and information from model updates. Machine unlearning has motivated extensive work on leakage after deletion. Comparing models before and after an update can amplify membership inference [2], enable reconstruction of deleted data [1], or reveal properties of the forgotten class. Gao *et al*. formalize deletion inference under paired model access [6], and Hu *et al*. recover the feature and label information of unlearned data from the original and unlearned classifier alike [7]. Our work is therefore not the first to study deletion-target inference. The difference lies in the observable and its nuisance structure. Our auditor inspects neither classifier logits nor parameter updates. The attack rests on *paired generation*: candidate prompts are issued to both checkpoints under matched seeds and compared in representation space. This yields a $K \times K$ displacement geometry whose columns differ sharply in sensitivity to unrelated edits, a confound that our calibration estimates from disjoint edits.

Weight-space alternatives. When the auditor can read both checkpoints, the parameter difference is itself informative. Task-arithmetic work shows that such differences behave as directions in weight space encoding the edit that produced them [8], and for a closed-form cross-attention edit like UCE the delta is analytically tied to the target's text embedding, so projecting it onto candidate embeddings would rank them directly and at negligible cost. We do not pursue this: our auditor has query access only (Sec. 3.1). Weight-space attacks are strictly better resourced and should perform at least as well where weights and edit method are both known. We ask what remains recoverable when they are not.

Behavioral model comparison. Comparing behavior before and after an intervention is a general auditing strategy, which diffusion models complicate: outputs are high-dimensional and stochastic, so pixel differences carry little meaning and unmatched samples differ through sampling alone. We fix prompts and seeds and compare generated images through a pretrained visual encoder [10,13], turning generative comparison into a controlled paired experiment. The contribution is the calibration of that displacement against unrelated interventions, not the encoder.

3 Problem and Method

3.1 Threat Model

Let M be an original text-to-image diffusion model and $M_t = \mathcal{U}(M, t)$ an edited model from which target concept t has been erased. The auditor knows that t

belongs to a closed candidate set $\mathcal{C} = \{c_1, \ldots, c_K\}$, but not which candidate was targeted. In our primary threat model (Setting P) the auditor additionally holds a set of unrelated calibration erasures $\mathcal{B} = \{M_{b_1}, \ldots, M_{b_B}\}$ whose targets are disjoint from $\mathcal{C}$, used only to estimate candidate susceptibility. The goal is to rank $\mathcal{C}$, ideally placing t first. The primary experiment uses $K{=}10$ and $B{=}5$.

The auditor has *query* access only: it may sample from either checkpoint under prompts, seeds, and sampler settings of its choosing, but does not read parameters, gradients, or activations, and does not know which erasure method produced M_t. This restriction is the point of the setting. If both checkpoints can be read and the editing procedure is known, the parameter delta is a far more direct observable, and a weight-space auditor should be expected to do at least as well as ours. Query-only access is the weaker and more broadly instantiated assumption, covering hosted inference endpoints and served derivatives; our numbers are correspondingly a lower bound on leakage, not an upper one. The setting is bounded in other ways too: we do not study erased-model-only inference, open-world discovery, or an auditor without a plausible candidate set.

3.2 Paired Generative Displacement

For candidate c_j we fix a prompt template $q(c_j)$. Given seed s, the two models generate $x^{(s)}_{M,j} = G(M, q(c_j), s)$ and $x^{(s)}_{M_t,j} = G(M_t, q(c_j), s)$; prompt, seed, sampler, and all inference parameters are matched across the pair. With a frozen encoder $E(\cdot)$ (DINOv2), the paired displacement is

$$D(t,j) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \Big[1 - \cos\big(E(x^{(s)}_{M,j}),\ E(x^{(s)}_{M_t,j}) \big) \Big]. \qquad (1)$$

Over deletion targets these scores form a matrix D: row t is the erased model, column j the candidate probe. The naive attack predicts $\hat{t}_{\mathrm{raw}} = \arg\max_j D(t,j)$, which is reliable only if D is diagonally dominant.

3.3 The Susceptibility Confound

Suppose c_j changes strongly not only when it is erased but also when unrelated concepts are erased. Then $D(t,j)$ can be large for $t \neq j$. We call this tendency *candidate susceptibility*. It is heterogeneous: in the discovery set Van Gogh has the largest average displacement under unrelated rows, and on the holdout Seurat accounts for four of five raw false positives (Sec. 5.1). As an explanatory model we write

$$D(t,j) = \mu + R_t + C_j + S_{t,j} + \epsilon_{t,j}, \qquad (2)$$

with global mean μ, row (edit-destructiveness) effect R_t, column (susceptibility) effect C_j, target–candidate interaction $S_{t,j}$, and residual $\epsilon_{t,j}$. A *deletion scar* corresponds to positive target-specific excess $S_{t,t}$.

Both attacks rank candidates *within* a row, so μ and R_t are constant across the comparison and cannot affect the predicted target: however destructive an edit is, it shifts every entry of its own row equally. Only C_j varies across the quantities being compared, which is why it alone must be estimated and removed. Equation (2) is explanatory rather than a causal identification: the terms need not be independent, and subtractive calibration cannot recover $S_{t,j}$ exactly. It motivates a practical question, namely whether C_j can be estimated independently well enough to improve ranking.

3.4 Independent Susceptibility Calibration

Figure 1 summarizes the full procedure. For each calibration deletion $b \in \mathcal{B}$ we probe all evaluation candidates and compute $D(b, j)$. Because calibration targets are disjoint from $\mathcal{C}$, these rows contain no positive match for any c_j. We estimate

$$\widehat{C}_j^{\mathrm{cal}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} D(b, j), \qquad A(t, j) = D(t, j) - \widehat{C}_j^{\mathrm{cal}}, \qquad \hat{t} = \arg\max_j A(t, j). \tag{3}$$

The priors $\widehat{C}_j^{\mathrm{cal}}$ are frozen before the unknown deletions are scored. The score reads simply: a candidate ranks high only if it moves more under the unknown deletion than it typically moves under unrelated deletions. If the $\mathcal{B}$ rows have mean-zero interaction and their row effects average out, then $\mathbb{E}[D(b, j)] \approx \mu + C_j$, so subtracting $\widehat{C}_j^{\mathrm{cal}}$ removes the column bias while the diagonal signal $S_{t,t}$ remains. Finite-B estimation error scales as $1/B$, which predicts diminishing returns as B grows (Sec. 5.5).

A transductive alternative estimates susceptibility from the other evaluation rows. It performs better on our data (Sec. 7) but is set-dependent, since evaluation targets contribute to one another's priors. Setting P uses only unrelated calibration deletions, keeping method development separate from the generalization claim.

4 Experimental Protocol

Evidence hierarchy. Experiments follow three levels (Tab. 1). The *primary confirmatory* experiment uses SD1.5 [14] with ESD-x, ten blinded holdout targets, ten candidates, and five independent calibration erasures, with attack, encoder, and operating strength frozen before evaluation. Two *replications* use smaller pre-specified target sets: one changes the backbone to SDXL [12] under ESD-style erasure, the other changes the intervention to UCE while keeping SD1.5. *Robustness and explanatory* analyses examine prompt wording, calibration budget and composition, candidate-set size, hard negatives, cross-backbone transfer, and a CLIP suppression proxy. We do not pool the primary and replication experiments.

Splits and blinding. The discovery stage used five artists (Van Gogh, Monet, Picasso, Dali, Kahlo) to identify paired displacement as the signal, diagnose the

Table 1: Experimental design. Primary and replication protocols are reported separately; sample sizes are not pooled.

Role	Backbone / method	n	K	$\lvert\mathcal{B}\rvert$	Evidence tier
Primary confirmatory	SD1.5 / ESD-x	10	10	5	Tier 1
Backbone replication	SDXL / ESD-x	5	10	3	Tier 2
Method replication	SD1.5 / UCE	5	10	3	Tier 2

susceptibility confound, select the operating strength, and motivate calibration. These artists are excluded from the holdout pool and their accuracy is never treated as confirmatory. A preregistered pool of thirty artists was audited on the original model. Ten evaluation artists (Lichtenstein, Goya, Schiele, O'Keeffe, Seurat, Manet, Rousseau, Rothko, Klimt, Kandinsky) and five disjoint calibration artists (da Vinci, Miró, Degas, Hopper, Modigliani) were fixed before any evaluation erasure was run, and the frozen protocol and frozen sets were hashed. One earlier freeze attempt failed on a text-encoder API mismatch and was reissued before any unlearning was performed. Calibration models were erased and probed first to produce the priors. The ten evaluation models were then processed and all 10×10 attack scores computed from anonymous model identifiers. The resulting prediction file contains scores and ranks but no target labels, and its SHA-256 digest was recorded before labels were joined in a separate step. All ten evaluation models completed, none was removed after evaluation, and neither the frozen sets nor the calibration priors were modified afterwards.

Generation and encoder. Primary probes use the template "a painting in the style of {artist}", three fixed seeds, 20 denoising steps at guidance scale 7.5, and 512×512 resolution, with a frozen DINOv2 ViT-B/14 encoder. The erasure operating point is 100 ESD-x steps, fixed before holdout evaluation because DINOv2 pairwise AUC on the discovery set was already saturated there, which avoids selecting the strength after seeing holdout data.

Determinism and the stochastic floor. Because the attack compares generations from different checkpoints, we audited reproducibility. Rerunning a probe within a session and after reloading the model produced bitwise identical images (matching SHA-256 digests) and a mean absolute DINOv2 rerun displacement of 6.0×10^{-8}, which we treat as the stochastic floor. Calibration priors sit between 8.8×10^{5} and 3.9×10^{6} times above this floor. Reproducibility does not extend across separate execution environments: regenerating the same probe in a different run changed measured displacement by up to 0.015, comparable to the weakest scars we observe. We therefore never regenerate images for a checkpoint that has already been probed, and reuse the stored generations, so all paired comparisons in a matrix come from a single run.

Metrics and inference. We report top-k accuracy, MRR, mean rank, and pairwise AUC. Under random ranking with $K = 10$ we have $\mathbb{E}[\text{Acc@1}] = 0.1$ and $\mathbb{E}[\text{MRR}] = H_{10}/10 \approx 0.293$. Significance uses an exact permutation test whose null hypothesis is that target labels are exchangeable across the anonymous deletion models, enumerating all 10! permutations. The smallest attainable p-value is therefore $1/10! \approx 2.8\times10^{-7}$, and we report that bound when no permutation reaches the observed statistic. The test addresses whether the ranking could arise without target information; it does not model variability from resampling which ten artists were deleted, and Sec. 5.3 treats that question separately. We additionally measure a CLIP text-image suppression proxy, the drop in target-text/image alignment after erasure, treated as descriptive rather than as ground truth for semantic forgetting.

5 Results

5.1 Raw Displacement Contains Signal and a Strong Confound

The discovery displacement matrix is diagonally structured: target-conditioned generations move more than off-target ones (Fig. 2a). Raw ranking is nonetheless unreliable, placing only two of five discovery targets first, and the failures are structured rather than random. Some columns receive large values across many unrelated rows (Fig. 2b). Van Gogh is the clearest case, with the largest mean off-diagonal displacement, and it remains the most susceptible probe at every erasure strength from 25 to 200 steps. Subtracting each candidate's background displacement, estimated from the other targets, exposes the diagonal (Fig. 2c) and lifts discovery accuracy from 40% to 80%. The useful signal lies not in raw change but in change measured against candidate-specific background susceptibility. Erasure strength affects targets unevenly, and we report that sweep in the supplement without claiming a general relationship between strength and scar magnitude.

5.2 Independent Calibration Reveals a Stable Susceptibility Hierarchy

The five unrelated calibration erasures produce a 5×10 matrix over the holdout candidates, with priors ranging $0.052 \leq \widehat{C}_j^{\text{cal}} \leq 0.230$. Roy Lichtenstein is the most susceptible holdout candidate, followed by O'Keeffe, Seurat, and Klimt; Rothko is the least. The hierarchy does not replay discovery, since Van Gogh is absent from the holdout. The priors are stable across calibration composition: the population standard deviation across the five calibration rows averages 0.024, and leaving out any one calibration artist changes holdout MRR by at most 0.024 (Sec. 5.5). Every prior exceeds the stochastic floor by at least five orders of magnitude, so susceptibility is neither measurement noise nor the product of a single deletion.

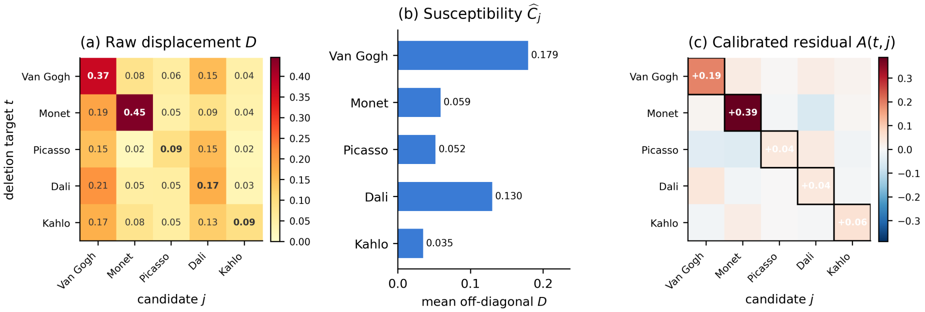

Fig. 2: Susceptibility confound on the discovery set (DINOv2, 100 ESD-x steps). (a) Raw displacement D has diagonal signal but strong columns. (b) Mean off-diagonal displacement per candidate: susceptibility is heterogeneous and largest for Van Gogh. (c) Leave-one-target-out calibrated residual $A(t, j)$; the diagonal (*outlined*) is positive for every discovery target, and four of five targets rank first. Dali is the exception.

5.3 Independent Calibration Generalizes to Blinded Holdout Artists

Table 2 and Fig. 3 summarize the primary result. Raw displacement already carries substantial information, ranking five of ten targets first with MRR 0.576 against a chance value of 0.293. Independent calibration raises top-1 accuracy from 50% to 70%, MRR to 0.814, and reduces mean rank from 3.8 to 1.8.

Two distinct questions arise, and the permutation test answers only the first. The calibrated MRR and the calibrated diagonal excess both exhaust the exact 10! enumeration: no relabelling of the ten sealed targets reproduces the observed statistic, so the test returns its smallest attainable value, $p = 1/10! \approx 2.76\times10^{-7}$. This is a floor rather than a measurement. It establishes that the calibrated ranking carries target information, and because the null is chance labelling, it says nothing about whether calibration improves on raw displacement. Calibrated Acc@1 gives $p = 1.3\times10^{-5}$ under the same null. Raw displacement, at 50% top-1, is already far from chance under any reading. The question that matters is therefore the comparison between the two attacks, not between either attack and chance, and we take it up below.

The gain is not a broad shift. Calibration improves five targets, leaves four unchanged, and worsens one (Fig. 3b). The largest corrections occur where raw ranking is attracted toward Seurat: Kandinsky improves from rank 6 to rank 1, Goya from 7 to 1, Schiele from 7 to 1, and Manet from 6 to 2. Seurat accounts for four of the five raw false positives, and subtracting its elevated background removes a systematic bias. Calibration does not merely amplify the diagonal; it changes which candidates the attack finds surprising.

Why calibration overcorrects. The primary failure is Roy Lichtenstein. It carries the largest susceptibility prior (0.230), yet it never once appears as a raw false positive. Seurat, whose prior is only third largest (0.194), produces four.

Table 2: Primary blinded holdout (n=10, K=10, Setting P). Chance baselines for K=10. Both attacks carry target information; the permutation test does not distinguish between them (Sec. 5.3).

Attack	Acc@1	Acc@3	MRR	AUC
Random	0.10	0.30	0.293	0.50
Raw DINO	0.50	0.50	0.576	0.71
Ind. calibrated	**0.70**	**0.90**	**0.814**	**0.88**

Exact 10! permutation on calibrated MRR attains the smallest attainable value, p=1/10!≈2.76×10^{-7}. This is a floor, not an effect size, and tests against chance rather than against raw displacement; the paired MRR gain over raw is +0.238, bootstrap [−0.088, +0.540].

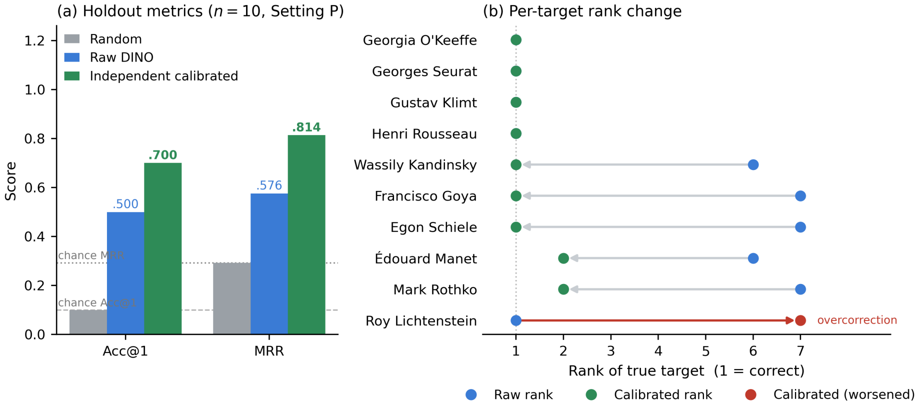

Fig. 3: Primary blinded holdout (n=10, Setting P). (a) Acc@1 and MRR for random, raw, and calibrated attacks; dashed and dotted lines mark chance (0.10, 0.293). (b) Per-target rank before (raw) and after (calibrated) calibration. Most targets improve or stay first; Lichtenstein regresses from rank 1 to rank 7 (overcorrection).

The prior therefore does not, on this holdout, identify the candidates that actually attract false positives; the two most extreme candidates on each quantity are different candidates, and with ten points the correlation between them cannot be estimated to any useful precision. Subtracting a large prior from a candidate that was never attracting false positives removes genuine evidence: Lichtenstein falls from rank 1 to rank 7 when it is the true target. Manet is difficult for a related reason, improving from rank 6 to rank 2 but never ranking first across the five prompts (Sec. 5.6). A fixed additive correction cannot distinguish susceptibility that generates false positives from susceptibility that is part of a genuine response.

What the permutation test does not establish. The permutation test asks whether the calibrated ranking could arise if scores carried no information about which target was deleted, and answers no. It does not compare calibration against raw displacement, and it does not model variability from resampling the ten deletion targets. Treating the targets as the sampling unit, calibrated Acc@1 is 0.70 with a 95% Wilson interval of $[0.40, 0.89]$, and the paired improvement in MRR over raw displacement is $+0.238$ with a bootstrap interval of $[-0.088, +0.540]$. A sign test over the six targets whose rank changed gives one-sided $p = 0.109$. The direction of the effect is consistent and the ranking is far from chance, but ten targets do not establish at conventional confidence that independent calibration beats raw displacement. We report the improvement as a description of this holdout, and regard a larger target set as the most valuable extension of this work.

5.4 The Signal Replicates Across Backbone and Erasure Method

We next ask whether deletion scars are specific to SD1.5 with ESD-x. On SDXL with ESD-x, the pre-specified five-target replication yields calibrated MRR 0.900 and 4/5 top-1 (p=0.008); on SD1.5 with UCE, calibrated MRR 0.669 and 3/5 top-1 (p=0.017). Neither is an independent n=10 confirmation, and with five targets the exact test enumerates only 120 relabellings, so no five-target study can report $p < 0.008$: both sit at or immediately above that floor, and the gap between these values and the primary p reflects enumeration size rather than effect size. The UCE result is weaker than both ESD-x results, so we do not claim scars are method-independent or equally strong across procedures. The narrower conclusion holds: target-specific paired displacement survives a change of backbone, and survives a change from iterative ESD-x fine-tuning to a substantially different cross-attention edit, though its geometry and strength vary by intervention.

Susceptibility itself transfers across backbones better than across methods: the frozen SD1.5 ESD-x priors applied to SDXL without recalibration recover all five targets, whereas on UCE transferred priors (0.651) match native ones (0.669) despite the two prior vectors being essentially uncorrelated ($r \approx -0.05$), so the subtraction is doing little work there. Full transfer diagnostics appear in the supplement.

5.5 Calibration Is Data-Efficient but Sensitive to Which Erasure Is Used

The frozen protocol uses five calibration models, which raises the question of whether so many are needed. Recomputing calibration retrospectively on subsets of the frozen five-row matrix, median MRR rises from 0.576 with no calibration to 0.728 with one erasure and 0.803 with three, reaching 0.814 at five: the median gain arrives immediately and saturates by three, consistent with the $1/B$ variance argument in Sec. 3.4.

The median hides a wide spread. With a single calibrator, MRR ranges from 0.659 to 0.914 across the five choices: the worst calibrator barely improves on no calibration at all, while the best exceeds the full five-erasure protocol. Which unrelated concept is erased therefore matters more than how many. The spread narrows quickly, and by $B{=}4$ leaving out any single calibrator moves MRR by at most 0.024. Auxiliary access remains necessary, since even the one-erasure attack needs an unrelated edited model. The full budget curve, composition, candidate-set size and hard-negative ablations are reported in the supplement.

5.6 Deletion Scars Persist Across Prompt Variation and Weak CLIP Suppression

We freeze four additional natural formulations and repeat the attack on the same ten deletion models. Calibrated MRR is 0.814 for the primary template "painting in the style of", and 0.682, 0.920, 0.733, 0.817 for "artwork in the style of", "painting created by", "artwork inspired by", and "characteristic style of" respectively; every prompt exceeds chance MRR (0.293), the mean is 0.793 at 68% top-1, and the mean across prompts likewise exhausts the permutation enumeration. The weakest prompt by MRR is the second and the lowest top-1 (50%) the fourth. These are repeated measurements on the same ten targets, not fifty independent deletions, and the five conditions are therefore strongly dependent. Robustness is target-dependent: Klimt ranks first under all five prompts; Kandinsky, Seurat, Rothko, Rousseau and Schiele under four; Manet never; and Lichtenstein under three, consistent with the overcorrection failure. Susceptibility rankings are stable across templates while calibration magnitude is only partly transferable: priors estimated on the primary template lose 0.03 MRR when applied to "characteristic style of" but 0.31 when applied to "painting created by". Per-prompt accuracies and mean ranks are tabulated in the supplement.

Three holdout targets have negative CLIP text-image erasure yet rank first after calibration: O'Keeffe (proxy -0.008, DINOv2 displacement 0.257), Klimt (-0.008, 0.353) and Schiele (-0.006, 0.110). The attack is not selecting the artist with the largest CLIP decrease. We read this as metric decoupling: paired representational displacement can remain informative when the CLIP suppression proxy is weak. It does not show that these artists were or were not forgotten.

6 Discussion

A *deletion scar* is not any change caused by editing. Erasure can globally alter a model, damage unrelated concepts, or disproportionately affect prompts that are naturally sensitive to parameter updates, so raw before-versus-after difference mixes several effects. We reserve the term for target-specific excess paired displacement, measured by $A(t,j) = D(t,j) - \widehat{C}_j^{\mathrm{cal}}$. The scar is behavioral: we do not claim it corresponds to a localized internal representation, a parameter subset, or a causal mechanism.

Calibration helps because the displacement matrix is not isotropic: a displacement of 0.20 is weak evidence for a candidate that usually moves 0.20 and strong evidence for one that usually moves 0.05, a distinction raw ranking cannot make. The Lichtenstein failure shows the limit of the idea. Susceptibility measured under unrelated erasures and susceptibility expressed as a false positive are related but distinct quantities, and a single subtractive prior conflates them, so it helps in aggregate and fails in particular. A more flexible model could estimate their interaction, or use the dispersion of the calibration rows rather than only their mean; we deliberately do not fit such a model on the holdout.

The replications establish external validity of the phenomenon, not universality. Both are five-target studies and UCE is clearly weaker, so the supported statement is that target-specific paired displacement replicates under a second backbone and a second erasure method with considerable method-dependent heterogeneity. On UCE in particular the calibration step contributes little, and the surviving signal is largely the raw displacement itself. Cross-backbone transfer hints that susceptibility partly reflects stable properties of the visual concepts, while weak cross-method transfer suggests that how a model is edited reshapes the nuisance geometry.

Most erasure evaluations ask whether the target was suppressed and whether utility was preserved. Our results add a third auditing question: what the pattern of change reveals about the intervention. This is not privacy leakage in every deployment, since our threat model is strong, requiring both checkpoints, a closed candidate set, and auxiliary erasures. Paired checkpoints are nonetheless plausible in open-weight research, model-version archives, and released edited derivatives.

Ethics. The concepts we recover are artist names from a public candidate set; no personal data or individual's identity is inferred anywhere in this work. In the deployments our threat model describes the erased target is typically announced rather than concealed, so the finding is not that a secret is exposed today but that a mechanism exists by which one could be, and the natural mitigation — releasing only the edited checkpoint — is cheap. All models here were edited by us from public checkpoints; we release the frozen protocol, score matrices, and analysis code, but no edited weights.

7 Limitations

The study is narrow by design, and the access assumptions are the strongest restriction. The auditor needs both M and M_t; we do not study erased-model-only inference. In the other direction, we restrict the auditor to query access and therefore say nothing about how much more a weight-space auditor could extract; our numbers are a lower bound on what is recoverable, not an upper one. The task ranks one target among ten known artists, so open-world discovery would require a different candidate-generation and statistical framework. Gain appears with one calibration erasure and saturates by three, but some auxiliary access remains necessary.

Scope is likewise limited. All targets are artist styles; objects and identities may have different susceptibility structure. We study SD1.5 and SDXL, ESD-x and UCE, and the weaker UCE result already shows that scar strength varies across interventions. The primary holdout contains ten independent deletions; the replications contain five each and are not pooled with it. DINOv2 is substantially stronger than CLIP image-image displacement here, so performance depends on the representation used. Displacements are averaged over three seeds, enough to stabilize the ranking on our data but too few to characterize the seed-level variance of D itself. The CLIP text-image change is a descriptive proxy: our weak-CLIP cases establish decoupling from that metric, not successful or failed semantic erasure.

The two failures share a cause. Neither Lichtenstein nor Manet is predicted by the size of the prior: the candidate with the largest prior attracts no false positives at all, while the candidate that attracts four has only the third largest. Subtraction is thus applied uniformly where it is not uniformly warranted. More expressive calibration might repair this — standardizing by the calibration-row dispersion, $(D(t,j)-\widehat{C}_j^{\mathrm{cal}})/\widehat{\sigma}_j$, is the obvious first attempt — at the cost of fitting choices we did not want to make on the holdout. A transductive leave-one-target-out variant does perform better, but it uses evaluation-set information, so we report it only as supplementary and never in place of Setting P. With ten targets the bootstrap interval on calibration's improvement over raw displacement includes zero; the ranking is far from chance under either attack, but the margin between them is not established at conventional confidence. A larger blinded target set is the single most valuable extension of this work, and the frozen protocol supports it without modification. We do not measure when the scar emerges along the denoising trajectory, nor whether internal activations give stronger evidence; public checkpoint collections as reusable calibration, erased-only threat models, and non-artist concepts all remain open.

8 Conclusion

We asked whether concept erasure leaves enough structure in a diffusion model's changed behavior to reveal which concept was targeted. Under our stated threat model it does, and it does so even before any correction is applied: on a frozen blinded holdout of ten untouched artist deletions, raw paired displacement ranks half the targets first and exhausts an exact permutation test. Raw displacement is nonetheless confounded, since some artist-conditioned generations are unusually sensitive to many unrelated erasures and become systematic false positives. Estimating that susceptibility from a disjoint set of unrelated erasures changes the attack from asking which candidate moved most to asking which moved more than expected, and on this holdout it improves top-1 accuracy from 50% to 70% and MRR from 0.576 to 0.814. Ten targets are too few to establish that margin at conventional confidence, and calibration fails instructively where it fails: the same prior that removes Seurat's false positives buries Lichtenstein when Lichtenstein is the true target.

The broader point is not that every erased concept is recoverable, but that model editing can create structured behavioral evidence about its own target, which stays partly hidden unless the comparison is calibrated.

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